

DUO ZHANG

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Education

Rutgers University

September 2023–Present

Ph.D. Candidate, Computer Science. Advisor: Prof. Jingjin Yu.

New York University

January 2021–December 2022

M.S., Courant Institute of Mathematical Sciences.

Shandong University

September 2016–June 2020

B.E., Computer Science (June 2018–June 2020); prior undergraduate study in Energy and Power Engineering (September 2016–June 2018).

Selected honor: National Scholarship, Ministry of Education of China.

Publications and Preprints

Conference and journal publications

- [1] **Duo Zhang**, Junshan Huang, and Jingjin Yu. “High-Performance Dual-Arm Task and Motion Planning for Tabletop Rearrangement.” *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. [Paper](#).
- [2] **Duo Zhang**, Zihe Ye, and Jingjin Yu. “Asymptotically-Optimal Multi-Query Path Planning for a Polygonal Robot.” *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. [Paper](#).
- [3] **Duo Zhang**, Chen Liang, Xifeng Gao, Kui Wu, and Zherong Pan. “Provably Feasible Semi-Infinite Program Under Collision Constraints via Subdivision.” *IEEE Transactions on Robotics (T-RO)*, 2024. [Paper](#).
- [4] Zherong Pan, **Duo Zhang**, Changhe Tu, and Xifeng Gao. “Planning of Power Grasps Using Infinite Program Under Complementary Constraints.” *IEEE Robotics and Automation Letters (RA-L)*, 7(1):650–657, 2022. [Paper](#).

Preprints and manuscripts

- [5] **Duo Zhang**, Hechen Zhang, Junshan Huang, and Jingjin Yu. “dRVG: Quadtree-Guided, Resolution-Complete Online Motion Planning for Polygonal Robots in Unknown Environments.” *arXiv preprint*, 2026. [arXiv:2609.31412](#).
- [6] **Duo Zhang**, Jintong Li, Junshan Huang, and Jingjin Yu. “ST-pRRTC: Parallel Space-Time RRT-C with Adaptive Goal-Time Forests.” *arXiv preprint*, 2026. [arXiv:2609.30533](#).
- [7] **Duo Zhang**, Zhizhuo Zhang, Xiaoli Bai, and Jingjin Yu. “Efficient Bézier Velocity Optimization for Free-Floating Space Manipulators.” *Manuscript*, 2026. FAVOR; arXiv and project links coming soon.

Research Experience

Rutgers University, Algorithmic Robotics & Control Lab

September 2023–Present

Graduate Assistant | Advisor: Prof. Jingjin Yu

- **Geometric and online motion planning.** Developed rotation-stacked visibility graphs for multi-query polygonal-robot planning (RVG), then extended the approach to unknown environments through local-roadmap merging and quadtree-guided sensing (dRVG). Evaluated against sampling-based planners and demonstrated online navigation on a microMVP robot. [2], [5]
- **GPU-accelerated manipulation and space-time planning.** Developed dependency-driven dual-arm rearrangement planning (SDAR) and adaptive goal-time forests for planning among moving obstacles (ST-pRRTC). Validated the methods on dual UR5e arms and a UR5e operating among Crazyflie quadrotors, respectively. [1], [6]
- **Free-floating space manipulation.** Developed Bézier joint-velocity optimization with analytical recursive sensitivities (FAVOR) for reaching, tracking, and prescribed-time interception under coupled spacecraft–arm motion. Evaluated in simulation with a seven-DoF arm and five spacecraft models. [7]

Tencent America, Lightspeed & Quantum Studio

June 2022–June 2023

Research & Development Intern, Copernicus Research Group

Bellevue, WA

- **Collision-constrained trajectory optimization.** Developed a semi-infinite programming solver using conservative motion bounds and adaptive subdivision to maintain collision-free trajectories; evaluated on articulated robot arms and UAVs. [3]

New York University, Courant Institute

September 2021–December 2022

Research Assistant with Profs. Daniele Panozzo and Lerrel Pinto

New York, NY

- **Deformable-object manipulation.** Built OpenAI Gym environments and PolyFEM tutorials for reinforcement learning with deformable objects; set up camera tracking and validated robot experiments, including a silicone-ball slingshot.

Shandong University, Interdisciplinary Research Center

January–December 2020

Research Assistant, Prof. Changhe Tu's lab

Qingdao, China

- **Power grasp planning.** Worked with Xifeng Gao and Zherong Pan on infinite programming with complementary constraints, kernel-integral relaxation, and fast multipole evaluation for optimization-based grasp planning. [4]

University of California, Los Angeles

July–September 2019

Research Assistant, Prof. Sriram Sankararaman's lab

Los Angeles, CA

Cross-disciplinary Scholars in Science and Technology Program

- **Cell-type-specific gene expression prediction.** Developed tensor-component-analysis methods for cell-type-specific gene-expression prediction and transcriptome-wide association studies; implemented and published an R package for the project.

Teaching Experience

Teaching Assistant, Rutgers University

- Software Methodology Fall 2024, Spring 2025
- Data Management for Data Science Spring 2024
- Systems Programming Fall 2023